Data Science for Mathematicians

Case Study: Predicting Bank Term Deposit Subscription

Department of Mathematics and Computer Science

Problem Statement

- Bank marketing campaigns aim to persuade clients to subscribe to **term deposits**.
- Identifying likely subscribers enables efficient resource allocation and reduces costs.

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- Bank marketing campaigns aim to persuade clients to subscribe to **term deposits**.
- Identifying likely subscribers enables efficient resource allocation and reduces costs.
- **Task**: Supervised binary classification.
- **Input**: Feature vector describing a client and the associated marketing campaign.
- **Output**: Whether the client will subscribe (`yes` / `no`).

Objectives

1. Clean the raw dataset and perform exploratory data analysis.
2. Engineer features via one-hot encoding; prepare a stratified train/test split.
3. Train seven classification models with hyperparameter tuning via 5-fold CV.
4. Evaluate all models on a held-out test set and select the best performer.
5. Deploy the chosen model as a serialised artifact for future inference.

Dataset Overview

- **Source**: Bank marketing dataset (Moro et al., 2014).
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age	job	marital	education	balance	housing	loan	deposit
59	admin.	married	secondary	2,343	yes	no	yes
56	admin.	married	secondary	45	no	no	yes
41	technician	married	secondary	1,270	yes	no	yes
55	services	married	secondary	2,476	yes	no	yes
54	admin.	married	tertiary	184	no	no	yes

Feature Description

Demographic

- `age` — client age
- `job` — type of job
- `marital` — marital status
- `education` — education level

Financial

- `default` — credit in default
- `balance` — yearly balance (EUR)
- `housing` — housing loan
- `loan` — personal loan

Campaign-related

- `contact` — communication type
- `day` — last contact day
- `month` — last contact month
- `duration` — call duration (s)
- `campaign` — contacts this campaign
- `pdays` — days since last contact
- `previous` — prior contacts
- `poutcome` — prior outcome

Descriptive Statistics

	age	balance	day	duration	campaign	pdays	previous
mean	41.23	1,528.54	15.66	371.99	2.51	51.33	0.83
std	11.91	3,225.41	8.42	347.13	2.72	108.76	2.29
min	18	−6,847	1	2	1	−1	0
max	95	81,204	31	3,881	63	854	58

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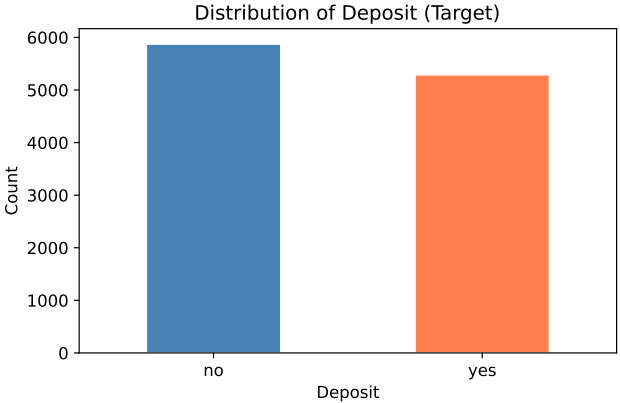
- Balance: large variability (std $\gg$ mean); some negative values.
- Duration: mean $\approx$ 372s ($\approx$ 6 minutes); heavily right-skewed.
- `pdays` $= -1$: client was not previously contacted.

Data Quality

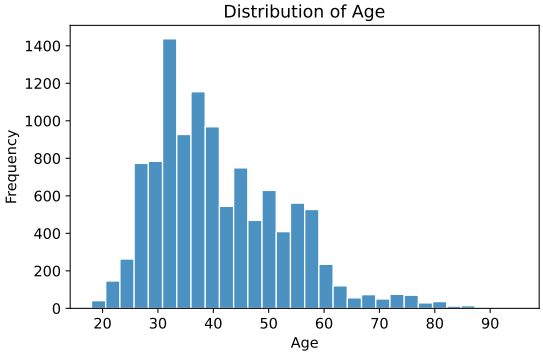
- **Missing values**: None across all 17 columns.
- **Duplicate rows**: None found.
- Dataset retained all 11,162 records without modification.

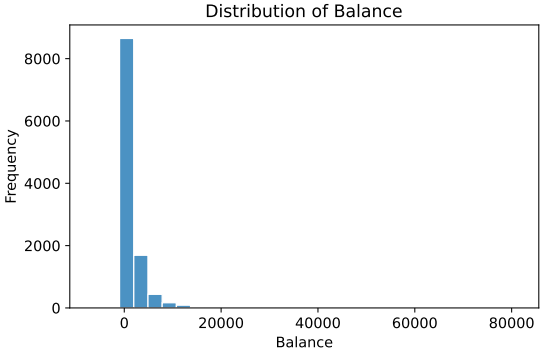
Target Distribution



- No: $n = 5{,}873$
- Yes: $n = 5{,}289$
- Class ratio: 0.90
- Roughly balanced — no resampling needed.

Numerical Features: Age and Balance

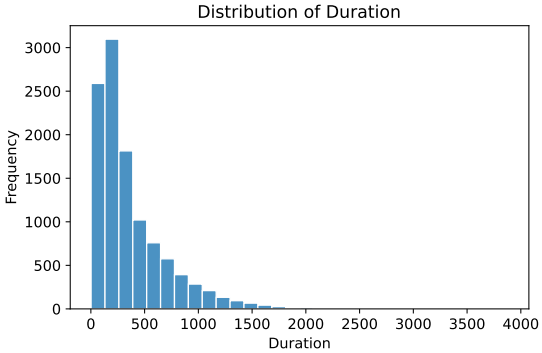


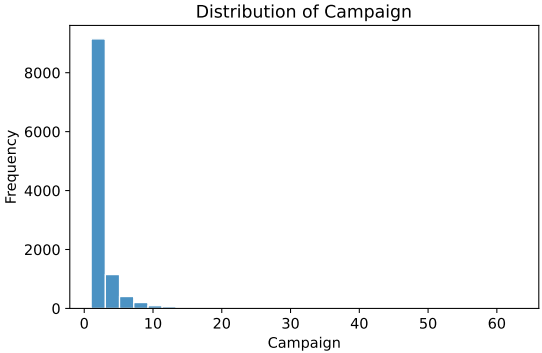


Age: slightly right-skewed, concentrated in 30–50 range.

Balance: heavily right-skewed with extreme outliers.

Numerical Features: Duration and Campaign

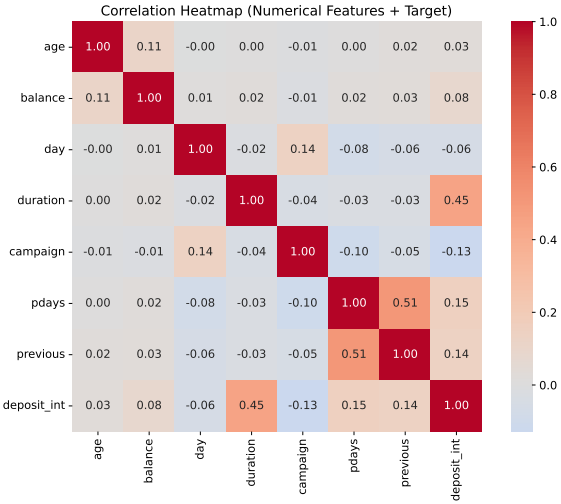




Duration: right-skewed, median = 255s.

Campaign contacts: most clients contacted 1–3 times.

Correlation Heatmap



Key Correlations with Target

Pearson correlation with deposit subscription (absolute value):

Feature	$\lvert r \rvert$
`duration`	0.45
`pdays`	0.15
`previous`	0.14
`campaign`	0.13
`balance`	0.08
`day`	0.06
`age`	0.03

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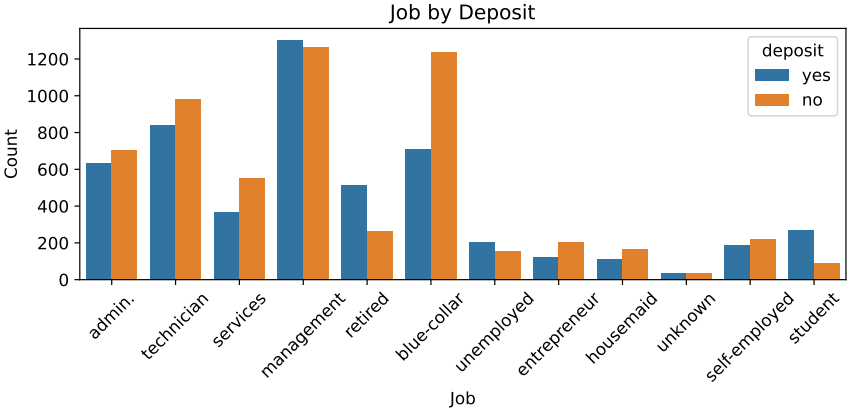
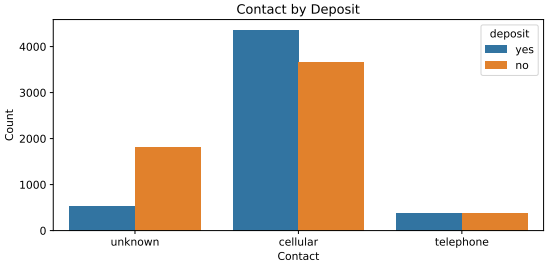
Takeaway: Call duration is the strongest linear predictor of subscription.

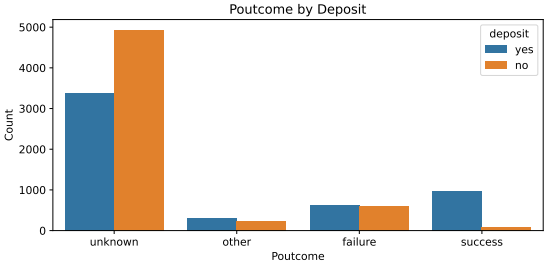
Categorical Features: Job Type



Students and retirees have higher subscription rates; blue-collar workers subscribe less.

Categorical Features: Contact and Previous Outcome

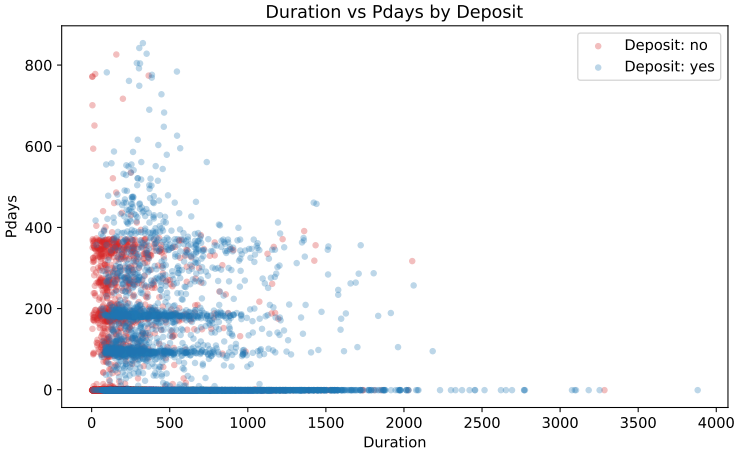




Cellular contact yields higher subscription rates.

Prior "success" strongly predicts subscription.

Top-2 Correlated Features: Duration vs Pdays



Longer call durations and positive pdays values are associated with subscription.

One-Hot Encoding

- Encode 9 categorical columns with `drop_first=True` to avoid multicollinearity.
- Target: 1 for "yes," 0 for "no."

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- Encode 9 categorical columns with `drop_first=True` to avoid multicollinearity.
- Target: 1 for "yes," 0 for "no."
- Result: feature matrix $\mathbf{X} \in \mathbb{R}^{11162 \times 42}$
  - 7 original numerical features
  - 35 binary indicators from 9 categorical variables

Stratified Train/Test Split

- **Split ratio**: 80% train / 20% test.
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Set	Total	No ($y = 0$)	Yes ($y = 1$)
Training	8,929	4,698	4,231
Test	2,233	1,175	1,058

Evaluation Metric: F_1 Score

Definition: F_β Score

The F_β score combines precision and recall:

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}}.$$

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- We use $\beta = 1$ (standard F_1), weighting precision and recall equally.
- Appropriate because the dataset is roughly balanced.
- Scoring: **weighted** F_1 via `GridSearchCV` with 5-fold CV.

Feature Scaling

- Apply `StandardScaler`: zero mean, unit variance.
- **Fit on training set only** to prevent information leakage.
- Transform both training and test sets.

$$x_j' = \frac{x_j - \hat{\mu}_j}{\hat{\sigma}_j}, \quad \hat{\mu}_j = \frac{1}{n_{\text{train}}} \sum_{i=1}^{n_{\text{train}}} x_{ij}, \quad \hat{\sigma}_j = \sqrt{\frac{1}{n_{\text{train}} - 1} \sum_{i=1}^{n_{\text{train}}} (x_{ij} - \hat{\mu}_j)^2}.$$

Training Pipeline

- Seven classification models, each tuned via `GridSearchCV`.
- 5-fold cross-validation with weighted F_1 as the scoring metric.
- All models receive the **scaled** features.

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Models trained:

1. Logistic regression
2. k-nearest neighbours (KNN)
3. Support vector classifier (SVC)
4. Decision tree
5. Random forest
6. Gradient boosting
7. Gaussian naive Bayes

Model 1: Logistic Regression

- Models $P(y = 1 \mid \mathbf{x})$ via the logistic function:

$$P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{x}^\top \boldsymbol{\beta}) = \frac{1}{1 + e^{-\mathbf{x}^\top \boldsymbol{\beta}}}.$$

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Hyperparameter grid

Parameter	Values
C	0.01, 0.1, 1, 10
penalty	none, ℓ_1, ℓ_2
solver	liblinear, saga

Best configuration

- $C = 0.1$, ℓ_1 penalty, liblinear
- CV $F_1 = 0.8233$

Model 2: k-Nearest Neighbours

- Prediction rule:

$$\hat{y}(\mathbf{x}) = \arg\max_{c} \sum_{i \in \mathcal{N}_k(\mathbf{x})} w_i \, \mathbf{1}[y_i = c].$$

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Hyperparameter grid

Parameter	Values
k	3, 5, 7, 11
weights	uniform, distance
metric	Euclidean, Manhattan

Best configuration

- $k = 11$, distance, Manhattan
- CV $F_1 = 0.7722$

Model 3: Support Vector Classifier

- Primal objective with kernel trick:

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n} \max\big(0,\, 1 - y_i(\mathbf{w}^\top \phi(\mathbf{x}_i) + b)\big).$$

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Hyperparameter grid

Parameter	Values
C	0.1, 1, 10
kernel	linear, RBF
γ	scale, auto

Best configuration

- $C = 1.0$, RBF kernel, $\gamma =$ auto
- CV $F_1 = 0.8396$

Model 4: Decision Tree

- At each node, select (j, t) minimising impurity:

$$\min_{j,t} \left[\frac{n_L}{n} \operatorname{Gini}(R_L) + \frac{n_R}{n} \operatorname{Gini}(R_R) \right], \quad \operatorname{Gini}(R) = 1 - \sum_c p_c^2.$$

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Hyperparameter grid

Parameter	Values
max_depth	3, 5, 7, 10
min_samples_split	2, 5, 10
criterion	Gini, entropy

Best configuration

- Gini, depth $= 7$, min_split $= 5$
- CV $F_1 = 0.8132$

Model 5: Random Forest

- Ensemble of B decorrelated decision trees:

$$\hat{y} = \text{majority vote}\big(T_1(\mathbf{x}),\ T_2(\mathbf{x}),\ \ldots,\ T_B(\mathbf{x})\big).$$

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Hyperparameter grid

Parameter	Values
B	100, 200
max_depth	5, 10, None
min_samples_leaf	1, 4

Best configuration

- $B = 200$, no depth limit, leaf $= 1$
- CV $F_1 = 0.8496$

Model 6: Gradient Boosting

- Sequential error correction:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta\, h_m(\mathbf{x}), \quad m = 1, \ldots, M.$$

Each h_m is fitted to the negative gradient of the loss.

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Hyperparameter grid

Parameter	Values
M	100, 200
η	0.05, 0.1
max_depth	3, 5

Best configuration

- $M = 200$, $\eta = 0.1$, depth $= 5$
- CV $F_1 = \boxed{0.8577}$ (highest)

Model 7: Gaussian Naive Bayes

- Assumes conditional independence given class:

$$P(\mathbf{x} \mid y = c) = \prod_{j=1}^{p} \frac{1}{\sqrt{2\pi\sigma_{jc}^2}} \exp\left(-\frac{(x_j - \mu_{jc})^2}{2\sigma_{jc}^2} \right).$$

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- **Best**: `var_smoothing` $= 10^{-9}$ (default).
- CV $F_1 = 0.6982$ — lowest among all models.
- The conditional independence assumption is violated by correlated features.

Test-Set Performance Summary

Model	Accuracy	F_1	ROC-AUC
Gradient Boosting	**0.8661**	**0.8662**	**0.9302**
Random Forest	0.8585	0.8586	0.9204
SVC	0.8491	0.8492	0.9187
Logistic Regression	0.8240	0.8238	0.9082
Decision Tree	0.8142	0.8141	0.8857
KNN	0.7815	0.7795	0.8696
Gaussian NB	0.7179	0.7079	0.8101

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Gradient boosting ranks first across all three test metrics.

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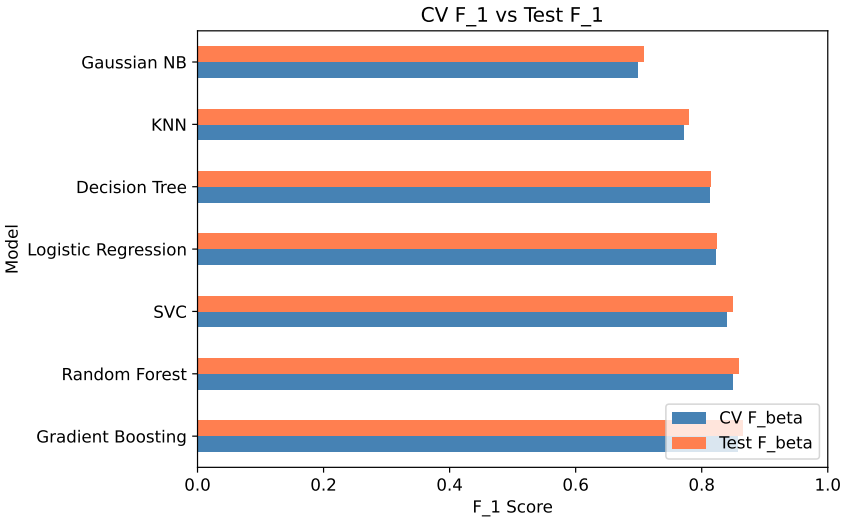
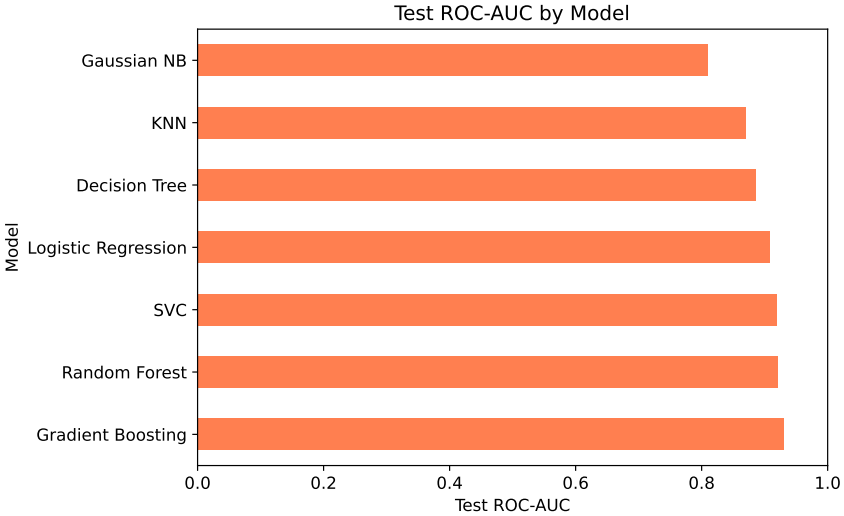
- Test F_1 is close to CV F_1 for all models — good generalisation.
- Ensemble methods (gradient boosting, random forest) outperform single models.

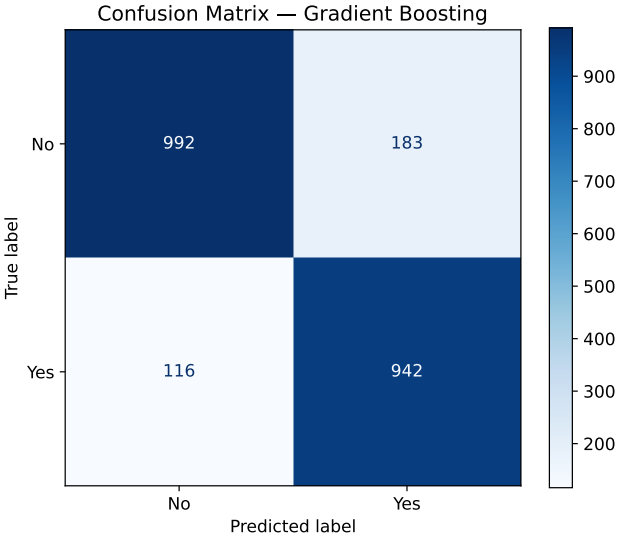
CV F_1 vs Test F_1



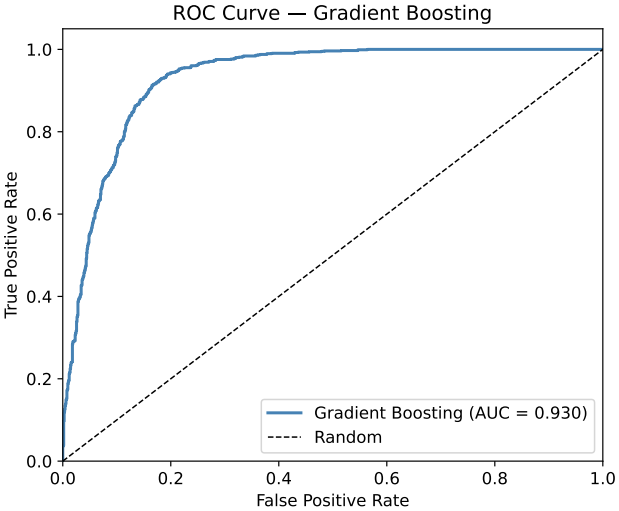
Test ROC-AUC by Model



Confusion Matrix — Gradient Boosting



ROC Curve — Gradient Boosting



AUC = 0.930: the model maintains high true positive rates even at low false positive rates

Serialised Model Artifact

The best model is saved as a `pickle` file containing:

Key	Description
`model`	Fitted `GradientBoostingClassifier`
`model_name`	`"Gradient Boosting"`
`scaler`	Fitted `StandardScaler`
`features`	Ordered list of 42 feature names
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Bundling ensures reproducible preprocessing and prediction without access to training data.

Inference on a New Observation

Example: Synthetic Client

A 35-year-old married technician with secondary education, balance of EUR 1,500, housing loan, no personal loan, contacted by cellular in June for 300 seconds.

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Pipeline: raw features $\xrightarrow{\text{scale}}$ scaled features $\xrightarrow{\text{predict}}$ class probability.

Output	Value
Predicted class	**yes**
$P(\text{no})$	0.1757
$P(\text{yes})$	**0.8243**

Summary of Findings

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- **Key predictors**: Call duration ($|r| = 0.45$) and previous campaign outcome.
- **Best model**: Gradient boosting — test $F_1 = 0.8662$, ROC-AUC $= 0.9302$.
- **Implication**: Longer, more engaged conversations and prior campaign success strongly predict term deposit subscription.

Limitations and Future Work

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- No macroeconomic features (interest rates, employment rate).
- `duration` is known only after a call ends — not a true pre-call predictor.
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Future directions

- Incorporate consumer confidence index and interbank lending rates.
- Explore target/ordinal encoding to reduce dimensionality.
- Test XGBoost/LightGBM with larger hyperparameter grids.
- Build duration-free models for true pre-call prediction.

Key Takeaways

1. The bank marketing dataset is clean and roughly balanced (0.90 class ratio).
2. Call duration is the strongest predictor of deposit subscription ($|r| = 0.45$).
3. Gradient boosting achieves the best test performance: $F_1 = 0.8662$, ROC-AUC $= 0.9302$.
4. Ensemble methods consistently outperform single models on this dataset.
5. The deployed model correctly classifies $\approx 87\%$ of unseen clients.
6. Feature engineering (one-hot encoding + standardisation) is essential for model comparability.